Zero-to-Hero: A Knowledge-guided Unsupervised Active Learning Pipeline for EEG-based Detection of Interictal Epileptiform Discharges [Extended Version]

Shen Liang
LIPADE, Université Paris Cité
Paris, France
shen.liang@u-paris.fr

Cihang Yu
ENSEA
Cergy, France
cihang.yu@ensea.fr

Qitong Wang
Harvard University
Cambridge, United States
qitong@seas.harvard.edu

Delphine Roussel
Institut Pasteur
Paris, France
delphine.roussel@pasteur.fr

Stephen Whitmarsh
Paris Brain Institute - ICM, APHP
Paris, France
stephen.whitmarsh@icm-institute.org

Katia Lehongre
Paris Brain Institute - ICM, APHP
Paris, France
katia.lehongre@icm-institute.org

Alexandra Levchenko
ISEP, LISITE
Issy-Les-Moulineux, France
alexandra.levchenko@isep.fr

Vincent Navarro
Paris Brain Institute - ICM, APHP
Paris, France
vincent.navarro@icm-institute.org

Themis Palpanas
LIPADE, Université Paris Cité
Paris, France
themis@mi.parisdescartes.fr

ABSTRACT

Interictal Epileptiform Discharges (IEDs) in EEG data are highly valuable for epilepsy diagnosis and treatment. Despite advances in deep learning-based IED detection, studies on reducing manual training data labeling are limited, especially on selecting the most informative examples to label, which is difficult due to data imbalance and heterogeneity. In response, we propose Zero-to-Hero (Z2H), an Unsupervised Active Learning (UAL) pipeline that starts from a *completely unlabeled* EEG dataset to select a high-quality set of examples under a very limited manual labeling budget. Z2H entails a knowledge-guided, data imbalance-aware initialization module, a data heterogeneity-aware active-PU transduction module and a knowledge-guided data filtering module for the best trade-off between data balance and heterogeneity. Z2H also features a GPU-accelerated approximation for the PU transduction process for time and memory efficiency. The data selected by Z2H can be used to both train a small model from scratch, and fine-tune an EEG Foundation Model (FM). Z2H can defeat multiple baselines while enabling (near-)real-time user interaction. When paired with an FM, Z2H can reduce manual labeling by 95% while having results similar to or better than manually labeling all available training data.

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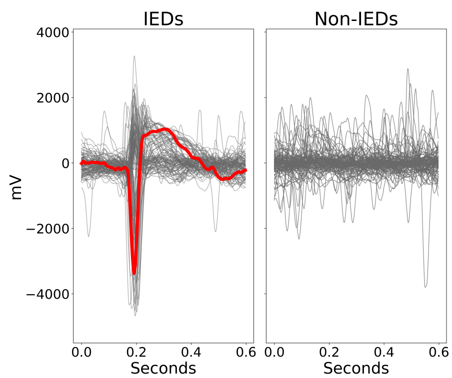


Figure 1: Butterfly plots of 100 IEDs (with one highlighted in red) and 100 Non-IEDs. IED detection can thus be formulated as a two-class classification problem between them. Note that here we only plotted a single channel for visual clarity, while *our method is designed with multi-channel data in mind*.

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/sliang11/Z2H/.

1 INTRODUCTION

Epilepsy, the second most burdensome neurological disorder [30], is characterized by recurrent seizures. In electroencephalography (EEG) data, fast, high-amplitude transients called *Interictal Epileptiform Discharges (IEDs)* [12, 39] (Fig. 1a) are highly linked to seizures [9] and thus valuable for epilepsy diagnosis and treatment [2, 39]. Recently, Deep Learning (DL) has been increasingly applied to IED detection, with most works formulating IED detection as two-class (IED/non-IED) classification [4, 10, 13, 15, 16, 34, 40, 43] (Fig. 1). Existing methods are typically fully supervised [13, 29, 34, 40, 43,

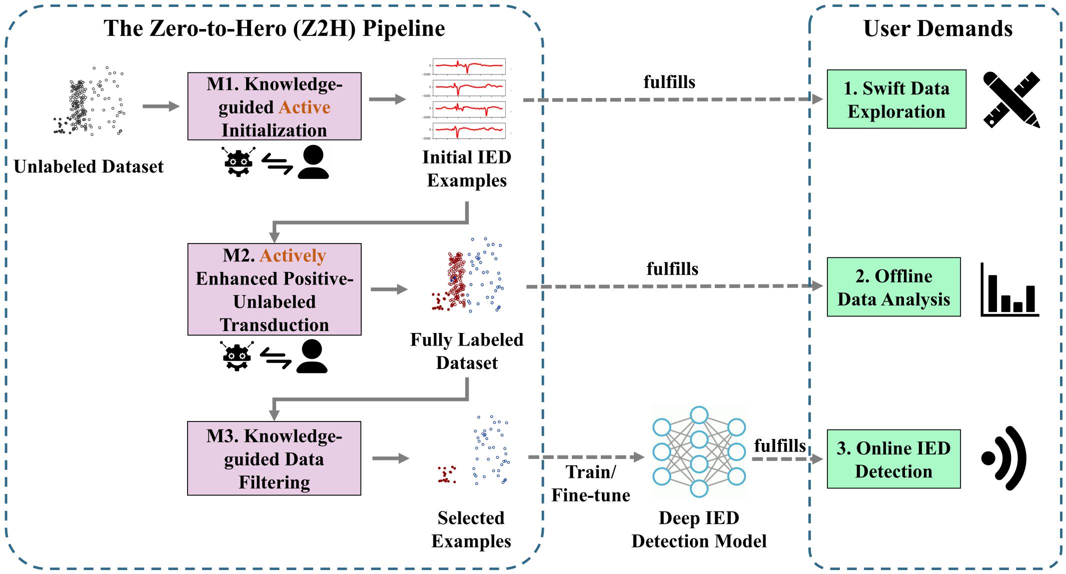


Figure 2: A summary of the proposed Zero-to-Hero (Z2H) pipeline. Starting with a *completely unlabeled* dataset, the pipeline uses 3 sequential modules to eventually select high-quality training/fine-tuning data for a deep model while obtaining intermediate outputs to meet diverse user demands.

49], targeting fixed, manually labeled training sets, which are very costly to produce. A few works explored reducing manual labeling by semi-supervision [13, 15] and data augmentation [15, 16], yet they can still require large numbers of pre-existing manual labels without considering an important question: *Given an unlabeled training set, which examples to select for manual labeling to minimize human effort while retaining high IED detection accuracy?*

In machine learning, this question translates to **Unsupervised Active Learning (UAL)**, a special case of Active Learning (AL) where the algorithm attempts to select a limited number (known as the AL *budget*) of most informative training examples for manual labeling, starting from a *completely unlabeled* dataset. To date, numerous UAL methods [8, 20, 22–25, 28, 31, 41] have been proposed, most of which [20, 22–25, 28, 31, 41] favor examples that can best reconstruct all or part of the dataset in a latent space. Existing methods are often limited in addressing two challenges prevalent in IED detection: 1) IEDs are very rare events, thus leading to significant **data imbalance** [43]. 2) Both the IED and non-IED classes can exhibit **data heterogeneity**. On the one hand, despite their relatively defined morphologies [46], IEDs can still be relatively diverse even for a single subject (e.g. a human patient), as shown in Fig. 1 *left*). On the other hand, the non-IED class is not well-defined and usually has a variance far greater than that of the IEDs.

Most existing UAL methods do not explicitly address these challenges, especially the methods that mainly preserve the global data structure [8, 24, 25, 31]. Some methods [20, 28] consider the local neighborhood of each example, which can only indirectly address these challenges by integrating more fine-grained information. Only a few methods [23, 41] explicitly handle these issues by preserving local clusters. However, as shown in Fig. 1 *right)*, the heterogeneity of the non-IEDs is usually characterized by highly random patterns, not by defined clusters. In fact, the data can often be so chaotic that we need to *increase the selected data size*, yet existing UAL methods can only select data for *manual* labeling without considering any examples that could be labeled *automatically* via semi-supervision. Thus, their selected data size is limited to the AL budget, which can often be small. While these methods can be fused with a separate semi-supervised learner proceeding them to expand the data size, the lack of *built-in* semi-supervision can still limit their effectiveness.

Another issue with existing UAL methods, in the context of IED detection, is that they may struggle to meet diverse user demands other than training the eventual deep IED detection model. In particular, the static training dataset itself may well be of interest to the user. For instance, suppose the dataset is derived from a 24-hour EEG recording for a human patient, the user may want to quickly examine a small number (e.g. 10-50) of IED examples in this dataset to explore the morphological characteristics of the IED events for this particular patient [46]; they may also want to study the frequencies of occurrences of IED events of this patient throughout the day. These demands do not necessitate the training of an online IED detection model primarily intended for unseen data [46]. It is therefore desirable to produce useful intermediate results to fulfill such user demands within the UAL process, which can be challenging for existing UAL methods.

Additionally, existing UAL methods typically suffer from scalability issues due to their extensive use of complex matrix operations that often have high time and space complexities. Specifically, traditional non-DL-based methods [11, 20, 24, 25, 31] usually have poor time efficiency. On the other hand, while DL-based methods [22, 23, 28, 41] can exploit GPU acceleration, they typically need to maintain $O(N^2)$ (N is the total number of examples) working memory to hold their scoring matrices, which greatly limits their utility in IED detection, where the data volume can often be large.

In this work, we propose **Zero-to-Hero** (**Z2H**, Fig. 2), a UAL pipeline that starts with an EEG dataset with zero labeled examples to eventually select informative labeled data to build a deep IED detection model (hence *zero to hero*). Specifically, Z2H entails 3 consecutive modules: 1) Addressing data imbalance, Z2H leverages a domain knowledge-guided active initialization module M1 intentionally biased to the IED class. 2) Based on M1, a Positive-Unlabeled (PU) [26, 45] transduction module, enhanced by a novel heterogeneity-aware AL-based post-processing method, is used to label *manually and automatically* the entire dataset, thus breaking the data size limit imposed by the AL budget. 3) Following M2, a knowledge-guided data filtering module filters likely mislabeled examples and enhances the trade-off between data balance and heterogeneity, yielding a high-quality set of selected examples. Besides the eventual output of Module M3, M1 and M2 produce their individual intermediate outputs to meet diverse user demands, which we will detail in Section 2. Replacing complex matrix operations with simple knowledge-guided heuristics and distance-based shallow learning, Z2H is far more lightweight than existing UAL methods. Z2H also features a GPU-accelerated approximation for PU transduction, which is its sole bottleneck.

In summary, our contributions in this work are as follows:

• We propose Z2H, which is, to the best of our knowledge, the first UAL method tailored to training/fine-tuning data selection for IED detection. With a knowledge-guided, active-PU methodology, Z2H aims to achieve an optimal trade-off between data balance, heterogeneity, and label correctness. Z2H is completely **model-agnostic** and the selected data can be used to build any *user-defined* IED detection model, which includes both training a small model from scratch and fine-tuning a pre-trained Foundation Model (FM). Utilizing 3 consecutive modules, Z2H produces diverse outputs to meet equally diverse user demands. Being lightweight and GPU-accelerated, Z2H can enable **(near-)real-time user interaction** with good time and memory efficiency.

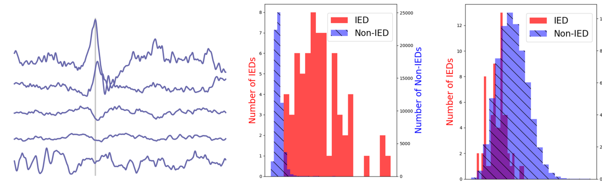
• Experiments show that Z2H can beat existing UAL methods in both *intra* and *inter*-subject settings. When paired with a pre-trained FM, Z2H can **reduce human labeling by 95%** and have similar or even better results than manually labeling all available examples, which can reduce the workload of neuroscientists from weeks to days or hours.

We note that unlike previous works, we are *not* proposing a new deep IED detection model. Rather, Z2H is a UAL pipeline to drastically reduce the manual labeling for training or fine-tuning *any* user-selected deep model without compromising accuracy.

For the rest of the paper, Section 2 details our Z2H. Section 3 presents the experiments. Section 4 reviews the related work. Section 5 concludes our work.

2 THE ZERO-TO-HERO (Z2H) PIPELINE

As mentioned above, Z2H entails 3 consecutive modules M1-M3, shown in Fig. 2. M1 involves a domain knowledge-guided heuristic AL sampling strategy that is *intentionally* biased to the IED class to obtain a set of initial IEDs. Based on these initial seeds, in M2, we draw on PU [26] transduction, a special case of one-class semi-supervised learning, to infer the labels of all unlabeled examples. These inferences are enhanced with a data heterogeneity-aware AL post-processing method. In M3, we again utilize domain knowledge



(a) An IED event and its non-IED context (IED Spike indicated by grey line) (b) Histogram of Peak-to-Peak (P2P) Amplitude (c) Histogram of Nearest Neighbor (NN) Distance

Figure 3: An illustration of the domain knowledge behind our Amp_{Dist}. (a) An IED event, with the IED spike present in only part of the channels, leading to higher amplitudes than the non-IED context. (b)(c) Histograms of P2P amplitudes and NN distances of all examples from a mouse, where IEDs generally have higher amplitudes and lower NN distances than non-IEDs.

to filter likely mis-inferred examples. The remaining examples, consisting of both manually labeled and automatically inferred ones, are selected to train or fine-tune the user-defined deep model. Besides the final selected examples obtained via M3, the intermediate outputs of M1 and M2 can fulfill other user demands than online IED detection (to be detailed later).

2.1 M1: Knowledge-guided Active Initialization

In M1, we use AL to query from the user the labels of a number of initial seed examples. We hope to *maximize the number of IEDs* among these seeds. In particular, we hope to *query examples that are likely to be IEDs first* for two reasons: On the one hand, we aim to *intentionally* bias the initialization process to the IED class to address data imbalance. On the other hand, we aim to meet the user demand of *swift data exploration*. Specifically, the user may wish to obtain a small number of IED examples with very few queries so that they can get a first impression of the characteristics of the IEDs in the dataset, such as delineating their rough morphologies [46], which requires the identification of *as many* IEDs as possible with the first few queries.

To do so, we exploit the following two pieces of **domain knowledge**: 1) IEDs are usually of high amplitudes [12], which suggests that IEDs tend to have higher *Peak-to-Peak (P2P) amplitudes* (the difference between the highest and lowest points) than non-IEDs (Fig. 3a, b). 2) Despite some heterogeneity, IED morphologies are generally defined by a sharp spike [12], habitually followed by slow waves [12, 39], which suggests that the IED space is likely more compact than the chaotic non-IED space. Thus, the *Nearest Neighbor (NN) distance* (here we use z-normalized Euclidean Distance) of an IED is unlikely to be very large (Fig. 3c).

We leverage K1 and K2 so that *examples more likely to be IEDs are queried first*, leading to a two-step $\boldsymbol{Amp_{Dist}}$ **sampling strategy**:

Step 1: *Obtain the Normalized Amplitude (NA) of each channel in each example, which is its P2P amplitude divided by the mean of the P2P amplitudes of that channel over all examples. Select the channel with the largest NA across all examples as the target channel.*

Step 2: *Using only the target channel, query the examples in P2P amplitude descending order, but skip those with NN distances larger than the median of the NN distances of all examples, and those whose NN example has already been queried.*

Here, we use only one channel rather than all channels, as IEDs spikes are usually only present in certain channels (Fig. 3a). Also, we use P2P amplitudes to decide the primary query order, while relegating NN distances to data filtering, as the former are more discriminative than the latter. This is shown in Fig. 3b and c, where the histograms of IED and non-IED P2P amplitudes are more separated than NN distances, likely due to the rarity and heterogeneity of IEDs diluting the density of the IED space. Moreover, we favor the median as the NN distance threshold over a more lenient one (e.g., $\mu + 3\sigma$ or $Q3 + 1.5 \times IQR$). This is a *precision-oriented* approach where we would rather filter atypically shaped IEDs than waste AL budget on high-amplitude noise.

Efficiency-wise, in interactive AL, we are most interested in the *user interaction response time* [27], i.e. the delay between two consecutive user queries. For Amp_{Dist}, by pre-computing and pre-sorting the amplitudes and distances, the response time is $O(1)$ as we can simply find the next example to query with simple lookups.

2.2 M2: Actively Enhanced PU Transduction

In M1, we can get a set PL of user-labeled positive (IED) examples and a set NL of negative (non-IED) ones. In M2, we label (manually and automatically) the set U of all remaining unlabeled examples to acquire more labeled data than the AL budget, for two reasons: On the one hand, we would like to augment the size of the labeled data to better address data heterogeneity. On the other hand, we hope to enable *offline data analysis*, where the fully labeled training set itself can be valuable to the user. For instance, for a 24-hour EEG recording of a patient, the user can use the fully labeled data to analyze the frequencies of IED occurrences at different times of the day.

We formulate the task of M2 as a one-class learning problem known as *Positive-Unlabeled (PU)* [26] transduction, which is to jointly use PL and U to label U. We favor this one-class solution over using both PL and NL examples because the Amp_{Dist} sampling strategy in M1 is intentionally biased to the positive class. Thus, NL cannot accurately represent the negative space. Concretely, we use the one-class **Self-Training 1-Nearest-Neighbor (ST-1NN)** [45] (Fig. 4) algorithm for PU transduction, which involves 2 phases:

ST-1NN Phase 1: Rank U examples by the possibility of being positive by initializing an ordered set R using PL examples, and iteratively moving the U example closest to R (measured by z-normalized Euclidean distance in our case) into R until U is empty.

ST-1NN Phase 2: Decide the decision boundary by a Stopping Criterion (SC) [7, 17, 36] estimating the number of positive examples np in R, labeling the first np examples positive and the rest negative.

Despite its simplicity, ST-1NN is generally suitable for use cases like IED detection where morphologically, the positive class is relatively compact while the negative class is not well-defined. However, it has two drawbacks: 1) ST-1NN alone often leads to sub-optimal results in IED detection. 2) ST-1NN can have poor space and time efficiency. We now detail how we address issues separately.

2.2.1 Active Enhancement of ST-1NN.. In IED detection, ST-1NN alone usually yields sub-optimal results for disregarding the heterogeneity of positive examples by drawing on a **Positive/Negative Conglomeration (PNC) assumption**: *in the ordered set R, the positive and negative examples conglomerate at the beginning and the end, respectively, with a single boundary between the two defined by the SC.* This is often not the case. For instance, in Fig. 4b, positive examples 11 and 12 are added to R *after* negative examples 8, 9 and 10, due to differences in the neighborhood around positive examples 6 (the NN of 11 in R when 11 was moved from U to R) and 7 (the NN of 8 and 9 in R), with 7 much closer to 8 and 9 than 6 to 11.

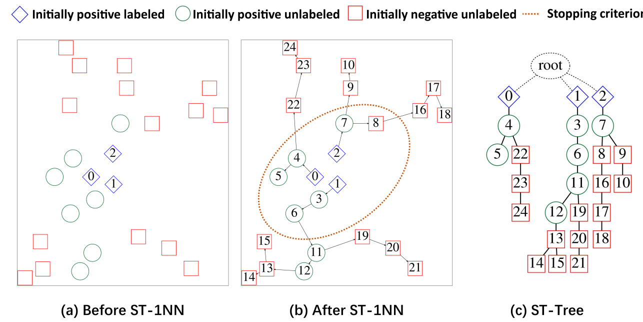


Figure 4: An illustration of ST-1NN. (a)(b) are the states before and after ST-1NN. The numbers in the nodes are in the order in which they are moved to the ordered set R. The edges link them to their NNs in R when they were added. The ellipse shows the decision boundary obtained by the stopping criterion. ST-1NN can be represented by a Self-Training Tree (ST-Tree) in (c).

In response, we draw on the *Self-Training Tree (ST-Tree)* [27] (Fig. 4c). Obtained by treating each example as a node with an edge between it and its NN in R, it naturally encodes data heterogeneity, with examples in different *branches* (paths from an initial PL example to a leaf node) likely diverging in morphology. Thus, while the PNC assumption does not hold well for the entire R set, the following **Branch-wise PNC (BPNC)** assumption holds much better: *each ST-Tree branch has 1) all positive nodes, or 2) both positive and negative ones conglomerated at the beginning and the end, separated by a branch-wise SC.* For instance, in Fig. 4c, the branch 0 — 4 — 5 is all positive, while 0 — 4 — 22 — 23 — 24 has positive examples 0, 4 at the beginning and negative ones 22, 23, 24 at the end. The branch-wise SC is thus between 4 and 22. Note that a branch cannot be all negative as it must start with a PL example.

While still an ideal assumption, BPNC is far more robust than PNC thanks to the reduced heterogeneity in each branch. Based on BPNC, we propose <u>A</u>ctive <u>S</u>topping <u>C</u>riterion <u>EN</u>hancement and <u>S</u>elect<u>ION</u> (**ASCENSION**), a novel AL-based ST-1NN post-processing method that both *selects* the best (non-branch-wise) SC from a pool of candidate SCs, and *enhances* its quality, in 2 steps:

ASCENSION Step 1: Active branch-wise inference. In this step (Alg. 1), we actively infer the labels of U examples in as many branches as possible (we will discuss the ordering of the branches later). For each branch, we first query from the user the ground truth label of its leaf node, leading to one of 2 cases.

Case 1: The leaf is positive (line 4 of Alg. 1). By BPNC, this branch is most likely all positive. For instance, for the branch 0 — 4 — 5 in Fig. 4c, by querying positive example 5, we infer 4 as positive.

Case 2: The leaf is negative (line 5-line 7). By BPNC, we find the branch-wise SC via an *active binary search*. Concretely, for each node checked, we query the ground truth labels of it and its parent: if both are positive, then we are in the positive space and need to

Algorithm 1: Active branch-wise inference (ASCENSION Step 1)

```
Input  : initial unlabeled examples U, an ST-Tree tree from ST-1NN Phase 1
Output : inferred positive and negative examples infP and infN
1 infP, infN = [], [];
2 while AL budget remains do
3     branch = nextBranch(tree); label = activeQuery(branch.leaf);
4     if isPositive(label) then infP.add(branch);
5     else
6         firstN, lastP = activeBinarySearchForBranchwiseSC(branch);
7         infP.add(branch.first : lastP);
          infN.add(tree.subTree(firstN));
8     tree.trim(infP, infN);
9 return infP, infN;
```

check the examples after this node; if both are negative, we check those before it. The search ends when we have found a negative node $firstN$ whose parent $lastP$ is positive, hence the branch-wise SC. For instance, in the branch 1 — 3 — 6 — 11 — 12 — 13 — 14, with the leaf 14 being negative, we query 11 and its parent 6, which are both positive. Thus, we then query 13 and 12, and find 13 as $firstN$. We label all examples before 13 as positive (e.g. 3), and all after as negative (e.g. 14). We also infer the descendants of $firstN$ in other branches (e.g. 15) as negative by BPNC.

Having checked the current branch, in line 8, we trim part of the sub-tree rooted at its initial PL example, removing all nodes in it with both themselves and all their descendants already labeled (queried or inferred). For instance, having labeled 1 — 3 — 6 — 11 — 12 — 13 — 14 (and 15), we can remove 12, 13, 14, 15, but keep 1, 3, 6, 11 as they have unlabeled descendants 19, 20, 21.

As for the order in which the branches are queried, we aim to maximize the number of examples inferred per query. Concretely, by finding $firstN$ in one branch, we can potentially trim the entire sub-tree $\mathcal{T}$ rooted at its initial PL example (if all ancestors of $firstN$ are in the branch). We denote the number of U nodes in $\mathcal{T}$ as u. Also, we need $O(lg(l))$ queries to find $firstN$, where l is the number of examples yet to be labeled in this branch. We thus score the branch by its estimated number of examples inferred per query, i.e. $u/lg(l)$. The branch with the highest score is queried next.

ASCENSION Step 2: SC selection. For the examples that remain unlabeled after Step 1 due to the limited AL budget, we use a pre-existing non-active, non-branch-wise SC [17, 45] to infer their labels. Concretely, we pre-select a pool of candidate SCs, each making its own inferences to the actively inferred examples in Step 1 in a non-active way. We use the actively inferred labels by ASCENSION as pseudo ground truths to assess the F1-score of each candidate. The best one is used to label all remaining unlabeled examples.

Looking back on ASCENSION, in Step 1, it essentially *enhances* all candidate SCs in Step 2 by replacing their non-active, non-branch-wise inferences with more accurate active and branch-wise ones. Moreover, the SC selection is done *after* the enhancement, not *before*, as the best SC before enhancement is not necessarily the one after. It is worth noting that a similar AL-based postprocessing method for ST-1NN was proposed in [27]. However, that method only aimed to select the best SC without enhancing it.

Efficiency-wise, the query order for ASCENSION needs adjustments on the fly with a worst-case user interaction response time of $O(|U|)$ with $|U|$ being the number of U examples. As we will show in Section 3.4, the response times are negligible in real applications.

2.2.2 GPU-accelerated Approximate ST-1NN. Despite being far more lightweight than existing UAL methods, Z2H is not without its bottleneck. Specifically, between the active interactions in M1 (Amp_{Dist}) and M2 (ASCENSION), the user must wait until ST-1NN is finished. We need to minimize the waiting time for a more seamless transition between M1 and M2. However, ST-1NN can entail a large number of pairwise distance calculations, which is $O(|PL||U|) + O(|U||U|)$ ($|PL|$ and $|U|$ are the initial sizes of PL and U) in the worst case[1]. It is thus preferable to pre-compute the distances before ST-1NN begins, yet this can lead to memory issues on large datasets. In response, we devise a simple ST-1NN approximation: for each PL and U example, we pre-compute its KNNs and the corresponding KNN distances among the other examples, reducing the number of distance to cache in memory to $O((|PL| + |U|)K)$ with K being a user-defined parameter. We consider the distance between each example with any example not among its KNNs as *infinity*, effectively disregarding such example pairs. Due to the rarity of IEDs (which means its $(K + i)$-NNs are not likely to be IEDs even when K is set to a modest number), this approximation can allow for results highly similar to those obtained using the exact ST-1NN with K set to a relatively small number.

However, even with pre-computation of the distances, ST-1NN can still be slow, especially on large datasets that require many iterations in Phase 1 of ST-1NN. To this end, we devise a simple GPU acceleration method, intended to reduce the per-iteration running time. Specifically, in each iteration, we need to find the smallest value from the pre-computed KNN distances between all (PL, U) pairs to find the next example to move from u to PL. If we reset the distance values of all non-(PL, U) pairs to greater than the largest among the (PL, U) pairs, we can convert the problem to finding the global minimum among all distance values, which is efficiently solvable on the GPU. As the iterations progress, we can restore the reset values to the original values for all non-(PL, U) pairs that turn into (PL, U) pairs, and reset the values for all (PL, U) pairs that turn into non-(PL, U) pairs in an incremental manner.

2.3 M3: Knowledge-guided Data Filtering

Despite the effectiveness of ASCENSION, the inferred labels in M2 can still contain mistakes that may affect the prediction accuracy of the eventual user-defined deep model (which we call the **core model**). Thus, in M3, we filter the fully labeled dataset obtained in M2.

Specifically, besides the *Actively Queried Examples (AQEs)* in M1 and M2, which have manual labels immune to mistakes, the training set entails 1) *Non-actively Inferred Examples (NIEs)*, which are *not* actively inferred in Step 1 of ASCENSION due to a limited budget, but inferred by the non-active SC in Step 2; 2) *Actively Inferred Examples (AIEs)*, which are inferred based on the labels of the AQEs in Step 1. Despite being the best candidate selected by ASCENSION, the non-active SC in Step 2 is still unlikely to outperform the active inferences in Step 1. Hence, *NIEs are more likely to be mislabeled than AIEs*. Thus, we impose different filtering policies on the two,

[1]While it is possible to reduce the number of raw distance calculation using some indexing or optimization method for high-dimensional 1NN queries [5, 19, 32, 33, 44, 47], such methods still require full distance calculations in the worst case.

Table 1: Data statistics. $|P|$ ($|N|$) is the number of IEDs (non-IEDs). Note that the training set is *initially unlabeled*, and all or part of it is later labeled via our Z2H or its rival UAL methods, before being used for core deep model training/fine-tuning. Unless an inter-subject setting is explicitly stated, all training data selection and core model training processes were conducted *independently* for each human/mouse subject, without referencing data from any other subject.

Dataset	Sampling Rate (Hz)	Length (seconds)	Number of Channels	Train (initially unlabeled)			Test			Data Description and Accessibility
				|P|	|N|	|P|/|N|	|P|	|N|	|P|/|N|	
H1				1782	93788	0.019	357	18758	0.019	
H2				2769	95981	0.029	554	19197	0.029	
H3				10178	84477	0.120	2036	16896	0.121	24-hour EEG data from 7 human patients. All patients gave written informed
H4		1.5		3043	93137	0.033	609	18628	0.033	consent (Project C11-16 conducted by INSERM and approved by the local ethics
H5			5	5526	90297	0.061	1105	18060	0.061	committee, CPP Paris VI). Data available on reasonable request for research
H6	512			2561	87229	0.029	512	17446	0.029	purposes.
H7				8089	72915	0.111	1618	14583	0.111	
H8				2208	90339	0.024	442	18068	0.024	
MO1				2639	56048	0.047	528	11210	0.047	1-hour EEG from 3 mice. All experimental procedures conformed to the European
MO2		0.09		394	56666	0.007	79	11334	0.007	Committee Council Directive (2010/63/EU) and were approved by the French
MO3			4	66	66110	0.001	14	13222	0.001	Ministry for Research and the local Ethical Committee. **Data made public at [1].**

leading to the following **Knowledge-guided Training Example Filtering (KTEF):** *1) Filter all NIEs. 2) Filter those AIEs inferred as IEDs (or non-IEDs), but with P2P amplitudes below (or above) the median P2P amplitude of all examples.*

Concretely, we remove *all* **NIEs** due to their lower confidence. For **AIEs**, we again draw on the domain knowledge that IEDs tend to have higher amplitudes (see Section 2.1), filtering the AIEs contradicting it. Similar to the NN distance threshold for Amp_{Dist} in M1, here we prefer the median amplitude over a more lenient threshold as a precision-oriented approach, namely, we would rather filter some correctly labeled examples than keep many incorrect ones. The data after filtering is used to train or fine-tune the core model.

As simple as it is, KTEF can yield high-quality selected data for the following three reasons: 1) Its emphasis on precision (over recall) can drastically enhance the **label correctness** in the selected data. 2) It does not remove all non-manually labeled examples, preserving the ability to break the **data size** limit by the AL budget. 3) It can enhance **data balance** in the selected data, as it can preserve a much higher proportion of IEDs than non-IEDs. This is because most IEDs have high amplitudes, while the amplitudes of non-IEDs are distributed more evenly, making the latter more likely to be filtered. We will empirically validate these in Section 3.1. Efficiency-wise, with pre-computed amplitudes and distances, KTEF takes $O(N)$ time as we simply need to conduct lookups for each example.

3 EXPERIMENTAL EVALUATION

We evaluate Z2H using the EEG data obtained from 11 subjects at the Paris Brain Institute (ICM), divided into the following two groups:

1) H1-H8: Intracranial stereoelectroencephalography (SEEG) data acquired from 8 human patients with refractory focal epilepsy who underwent a presurgical intracerebral electrode investigation in the epilepsy unit of a hospital. All intracranial SEEG procedures were performed according to clinical practice and the epilepsy features of the patients. All patients gave written informed consent (Project C11-16 conducted by INSERM and approved by the local ethics committee, CPP Paris VI). Each patient was implanted with intracranial depth electrodes with 4 to 8 macroelectrode contacts separated by 5 mm (1.3 mmØ). The trajectories, anatomical targets, number of electrodes as well as electrode contacts, were determined as per the clinical practice and the epilepsy features of the subjects. The raw recording of each subject was continuously recorded at 4096 Hz with a high-pass filter at 0.01 Hz. IED events in the first 24 hours of each recording were manually annotated as per standard guidelines [18]. For data preprocessing, we followed the practice of [43], which involves bipolar-referencing the data, applying a bandpass filter of 1-50 Hz, downsampling to 512 Hz, and using 1.5s sliding windows to segment the continuous recordings into examples. This resulted in training and testing examples with the length of $512 \times 1.5 = 768$.

2) MO1-MO3: EEG data acquired from 3 epileptic mice. All experimental procedures were conducted in compliance with the European Committee Council Directive (2010/63/EU) and received approval from the French Ministry for Research and the local Ethical Committee. The data involves 2 types of epileptic models: 1) MO1 had *induced* epilepsy. Concretely, it was anesthetized with 2-4% isoflurane, received analgesia (Buprecare, 0.1 mg/kg), and was positioned in a stereotaxic frame. Kainic Acid (KA, 50 nL; 20 mM in PBS) was injected into the right hippocampus at coordinates relative to the bregma (antero-posterior, -1.8 mm; medio-lateral, -1.8 mm; dorso-ventral, -1.8 mm). The mouse was then implanted with 2 bipolar stainless steel electrodes in the right hippocampus, and 3 other stainless steel electrodes in the right and left primary motor cortexes, as well as on the cerebellum as a reference. 2) MO2 and MO3 had spontaneous epilepsy for *genetic* reasons, for which stainless steel electrodes were implanted on the left and right primary motor cortexes, the left somatosensory cortex, the left and right lateral parietal association cortexes, and in the cerebellum as a common reference. The three mice were connected to an amplifier as part of a video-EEG acquisition system. Signals were acquired at 4096 Hz and band-pass filtered ($0.5 - 70$ Hz). Continuous video-EEG recordings were conducted with mice having unrestricted access to food and water throughout the recording sessions. The first-hour recording was selected where the annotations were made using the EDFbrowser software[2]. For data preprocessing, we followed a similar protocol to that for the human data H1-H8, again applying a bandpass filter of 1-50 Hz, downsampling to 512 Hz, and using sliding windows to segment the continuous recordings into examples. As with the window size, we favored 0.09s windows over the 1.5s ones for the

[2] https://www.teuniz.net/edfbrowser/

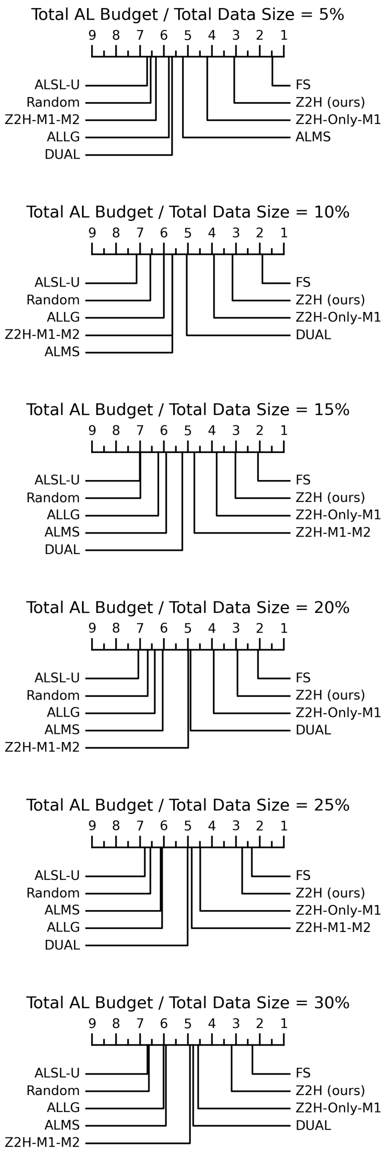


Figure 5: CD diagrams of Z2H, its ablations and rival UAL methods. *FS* refers to fully supervised training with all available data. Methods to the right perform better.

Table 2: Online prediction F1-scores of core deep learning models on the testing sets when the total AL budget is 5% of all data, using Z2H, its ablations, and other rival UAL methods for training (or fine-tuning, in the case of pre-trained *Foundation Models (FM)*) data selection. The best results (except for *Fully Supervised (FS)*) are shown in bold, and second bests underlined. "-" indicates failed training or invalid fields.

Core Model	Method	H1	H2	H3	H4	H5	H6	H7	H8	MO1	MO2	MO3	Avg. F1	Avg. Rank (w/o FS)
AIED	*FS*	0.66	0.59	0.83	0.70	0.87	0.81	0.90	0.75	0.92	0.96	-	0.80	-
	Random	0.51	-	0.72	0.45	0.79	0.55	0.82	0.60	0.86	-	-	0.66	5.83
	ALSL-U	0.55	-	0.73	-	0.75	0.56	0.47	-	0.84	-	-	0.65	6.00
	ALLG	0.60	-	**0.74**	0.49	**0.82**	0.58	0.84	0.67	0.87	-	-	0.70	2.17
	ALMS	0.57	-	0.73	0.46	0.79	0.35	0.84	0.71	**0.88**	-	-	0.67	4.17
	DUAL	0.47	0.10	0.73	0.45	0.81	0.61	0.78	0.65	0.83	**0.94**	-	0.64	5.17
	Z2H-Only-M1	0.60	0.25	**0.74**	0.53	0.80	0.64	0.83	0.70	0.81	0.91	-	0.68	3.50
	Z2H-M1-M2	**0.63**	0.39	0.63	0.58	0.68	0.65	0.72	0.70	0.72	0.89	-	0.66	5.67
	Z2H (ours)	0.58	**0.43**	**0.74**	**0.66**	**0.82**	**0.77**	**0.86**	**0.73**	0.87	**0.94**	**0.78**	**0.74**	**1.67**
iEDeal	*FS*	0.84	0.75	0.93	0.87	0.94	0.93	0.96	0.88	0.91	0.79	0.60	0.85	-
	Random	0.56	0.39	0.75	0.58	0.78	0.61	0.84	0.71	0.87	0.59	0.47	0.65	4.80
	ALSL-U	0.55	0.22	0.75	0.46	**0.80**	0.63	0.84	0.71	0.86	0.76	-	0.66	5.10
	ALLG	0.54	0.42	0.74	0.55	**0.80**	0.61	0.85	0.72	0.87	0.21	-	0.63	4.70
	ALMS	0.57	0.42	0.73	0.58	0.79	0.62	0.86	0.70	0.86	0.19	-	0.63	5.10
	DUAL	0.58	0.39	0.75	0.55	0.77	0.59	0.84	0.69	**0.88**	0.80	0.58	0.67	5.00
	Z2H-Only-M1	0.63	0.43	0.77	0.62	**0.80**	0.65	0.84	0.75	0.87	0.81	**0.67**	0.71	2.50
	Z2H-M1-M2	**0.71**	**0.58**	0.68	**0.73**	0.74	**0.82**	0.78	0.79	0.71	0.82	0.50	0.71	4.00
	Z2H (ours)	0.61	0.37	**0.86**	0.60	0.79	0.70	**0.90**	**0.84**	**0.88**	**0.83**	0.61	**0.73**	**2.40**
IEDConformer	*FS*	0.66	0.46	0.80	0.70	0.84	0.74	0.88	0.77	0.92	0.89	0.59	0.75	-
	Random	0.56	0.25	0.74	0.53	0.78	0.64	0.83	0.69	0.88	0.76	0.36	0.64	5.18
	ALSL-U	0.59	0.31	0.75	0.21	0.78	0.65	0.82	0.66	0.88	0.58	0.33	0.60	5.55
	ALLG	0.61	0.42	0.75	0.52	0.77	0.64	0.82	0.74	0.89	0.62	0.67	0.68	4.18
	ALMS	0.60	0.19	0.77	0.51	**0.79**	0.70	0.83	**0.76**	0.89	**0.83**	0.50	0.67	3.55
	DUAL	0.55	0.23	0.75	0.54	**0.79**	0.58	0.81	0.70	0.90	**0.83**	0.54	0.66	4.64
	Z2H-Only-M1	0.62	**0.47**	**0.78**	**0.65**	0.77	**0.73**	**0.85**	0.75	**0.91**	0.76	0.64	**0.72**	**2.09**
	Z2H-M1-M2	0.62	0.27	0.56	0.58	0.70	0.62	0.74	0.68	0.71	**0.83**	0.60	0.63	5.55
	Z2H (ours)	**0.67**	0.45	0.75	0.61	0.78	**0.73**	0.82	0.73	0.87	0.72	**0.70**	0.71	3.09
EEGPT (FM)	*FS*	0.84	0.60	0.90	0.84	0.93	0.86	0.95	0.89	0.91	0.86	0.73	0.85	-
	Random	0.77	0.59	0.87	0.82	0.90	0.76	0.93	0.87	0.89	0.82	0.13	0.76	5.00
	ALSL-U	0.79	0.60	**0.88**	0.78	**0.91**	0.76	**0.94**	**0.88**	0.89	0.80	0.35	0.78	3.73
	ALLG	0.77	0.63	0.87	0.77	**0.91**	0.74	0.92	0.87	**0.90**	0.81	0.13	0.76	4.91
	ALMS	0.81	0.58	**0.88**	0.82	**0.91**	0.80	**0.94**	**0.88**	0.89	0.83	0.35	0.79	2.73
	DUAL	0.80	0.61	0.87	0.82	**0.91**	0.77	0.93	0.86	0.89	0.86	0.64	0.81	3.55
	Z2H-Only-M1	0.36	0.66	**0.88**	**0.84**	**0.91**	0.81	0.93	0.87	**0.90**	0.83	0.78	0.80	2.64
	Z2H-M1-M2	0.81	0.43	0.68	0.76	0.77	0.70	0.78	0.79	0.62	0.83	0.60	0.71	6.64
	Z2H (ours)	**0.82**	**0.70**	0.87	**0.84**	**0.91**	**0.83**	**0.94**	**0.88**	0.88	**0.88**	**0.83**	**0.85**	**1.82**
NeuroGPT (FM)	*FS*	0.87	0.71	0.92	0.87	0.94	0.93	0.95	0.89	0.86	0.94	0.25	0.83	-
	Random	0.82	0.65	**0.90**	0.82	0.91	0.89	0.93	0.84	0.76	0.57	0.02	0.74	4.30
	ALSL-U	0.83	0.65	0.89	0.79	0.90	0.89	0.92	0.83	0.76	0.37	0.08	0.72	5.30
	ALLG	0.83	0.64	0.89	0.82	0.91	0.90	0.92	0.86	0.78	0.59	0.11	0.75	3.70
	ALMS	0.83	0.62	**0.90**	0.83	0.91	0.90	0.93	0.86	0.78	0.49	-	0.80	3.50
	DUAL	0.83	0.65	0.89	0.82	**0.92**	0.91	0.93	0.86	0.49	0.73	0.24	0.75	3.30
	Z2H-Only-M1	0.80	0.64	0.89	**0.85**	0.90	**0.93**	**0.94**	0.86	0.58	0.76	0.36	0.77	3.70
	Z2H-M1-M2	**0.86**	0.60	0.70	0.75	0.67	0.82	0.79	0.86	0.64	**0.91**	0.13	0.70	5.80
	Z2H (ours)	0.85	**0.71**	0.89	**0.85**	**0.92**	**0.93**	**0.94**	**0.88**	**0.81**	0.87	**0.55**	**0.84**	**1.40**

human data, as the durations of IEDs in the mice are much shorter than those in the humans. We also let the windows have 0.045s (i.e. half the window size) overlaps. This is because, unlike the 24-hour human data, we only have one hour of data per mouse. Thus, overlaps are needed to populate the data.

The data was split into training and testing sets with a 4 : 1 ratio. Table 1 shows the data statistics. Unless otherwise noted, all experiments were run *independently* for each subject, without referencing data from any other subject. Although in Section 3.1, we will evaluate Z2H in an inter-subject setting. **The mouse data is openly accessible at [1].**

Being model-agnostic, Z2H can be fused with any user-defined core model, for which we consider the following two groups:

1) **AIED** [34], **iEDeal** [43], and **IEDConformer** [35]: small models trained from scratch with data selected by Z2H, with AIED and iEDeal being CNNs and IEDConformer embracing a hybrid of convolutional and Transformer-based architectures. In terms of hyper-parameter settings, for AIED, we used the original authors' settings. For iEDeal, the default settings were mostly used. However, we made two modifications: First, apart from SaSu, we also considered the conventional Binary Cross-Entropy (BCE) loss, as SaSu can sometimes lead to convergence failures for the mouse data (although such failures

can occur for BCE as well, but less frequently than SaSu). Second, the original authors used $\{0.005, 0.01, 0.02\}$ as learning rates. However, we found that these learning rates can often lead to sub-optimal results for the mouse data. Thus, we expand the range of possible learning rates to $\{0.001, 0.002, 0.003, 0.005, 0.01, 0.02, 0.03, 0.05\}$. For IEDConformer, we followed all the hyperparameter settings the authors specified. Unfortunately, the authors did not specify important hyperparameters such as the learning rate, the mini-batch size and the number of training epochs. For these hyperparameters, through a grid search, we found that a learning rate of 0.001, a mini-batch size of 64 and 200 epochs generally lead to good performance, which is the setting we used.

2) **EEGPT** [42] and **NeuroGPT** [11], which are pre-trained EEG Foundation Models (FMs) using Z2H for fine-tuning data selection. For both FMs, we mostly followed the original authors' settings. In particular, for NeuroGPT, we used the *encoder-only* paradigm of fine-tuning, as this led to the best results in the original authors' experiments. An important adaptation we made for both FMs is that they require the input examples to have a fixed number of channels (58 for EEGPT, 22 for NeuroGPT) and length (1024 for EEGPT, 500 for NeuroGPT). To adapt our data to such criteria, we used a learnable 1D convolutional layer to map the channels of our data to the required ones, and used per-channel linear interpolation to resample our data to meet the length requirements.

For all the aforementioned core models, following the practice of [43], we used a $4:1$ ratio to further split the training data into fitting and validation sets. While it is unconventional to use validation sets that may contain incorrect labels, we noticed that some core models (especially iEDeal) are highly sensitive to hyperparameter settings (especially the learning rate), which is why we made this decision.

Note again that Z2H is *not* an IED detection model, but a UAL method intended to select training examples for a separate core model. Thus, the core models above are *not* baselines competing with Z2H.

We ran our experiments with an Intel Xeon Platinum 8468 CPU, an NVIDIA H100 GPU and 512 GB RAM. All code [1] is in Python 3.11, with the GPU-based methods implemented with PyTorch 2.4.0.

For the rest of this section, we will first compare our Z2H with rival UAL methods, as well as its ablations. We will then examine Z2H's ability to fulfill diverse user demands by assessing the quality of the intermediate results produced by M1 and M2. Finally, we will evaluate the user interaction efficiency of Z2H.

3.1 Comparison with Baselines and Ablations

We compare Z2H with the following baseline and ablation methods.

- **Fully-Supervised (FS):** manually label *all* available examples. A *theoretical upper bound*, it can be *practically* beaten sometimes.
- **Random:** a random sampling strategy.
- **ALSL-U [24], ALLG [28], ALMS [22], DUAL [23]:** Several state-of-the-art UAL baselines.
- **Z2H-Only-M1, Z2H-M1-M2:** Z2H ablations; the former only selects examples manually labeled in M1, the latter selects the entire dataset labeled in M2 while omitting data filtering in M3.

We set the total AL budget to $5\% : 5\% : 30\%$ of all training data. For Z2H-M1-M2 and Z2H, we set the number of nearest neighbors

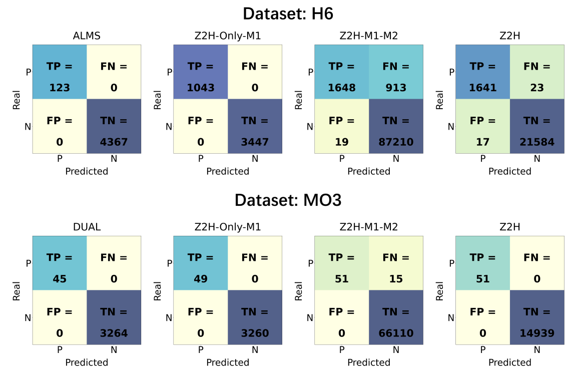


Figure 6: Confusion matrices of the selected training data. Note that for methods other than Z2H and Z2H-M1-M2, the labels are always correct as all selected examples are manually labeled.

K for approximate ST-1NN in Module M2 to 1,000, which allows for highly similar results as the exact ST-1NN while being time and memory-efficient. For the candidates SCs in M2, we use GBTRM [17], a *family* of 5 SCs. For each of the 5 SCs, we set 5 candidate parameters following the practice of [27]. Specifically, each of the 5 SCs entails a parameter named β, for which we choose from 0.1, 0.2, 0.3, 0.4, 0.5. This leads to a total of $5 \times 5 = 25$ candidates. Also, we let Modules M1 and M2 share a single total budget, with a $7:3$ split between the two. We will justify this budget allocation in Section 3.3. For Z2H-Only-M1, we reallocate the AL budget intended for M2 in Z2H-M1-M2 and Z2H to M1 for fair comparison. For the other methods, we mostly preserve the parameter settings of the original authors, with minor tweaks for certain methods to better adapt to our data. In particular, for ALMS, we changed the learning rate from 0.0005 to 0.001 as this led to better convergence on our data. All results were averaged over 10 runs.

We compare the methods above by pairing them with the core deep models mentioned earlier and evaluate them by their online prediction F1-scores on the testing sets. Due to page limits, we cannot present all raw results (available at [1]) here. Instead, we draw on the widely used [6, 14, 27] Critical Difference (CD) diagrams to compare their statistical differences, shown in Fig. 5. Specifically, for each total AL budget, we compare the F1-scores of all methods obtained on all combinations of the 5 core models and 11 human/mouse subjects using a combination of the Friedman test and the Wilcoxon signed rank test. In each CD diagram, methods to the right rank higher on average. Should two or more methods have no significant statistical difference, they would be grouped in the same clique (not present in our results). Apart from the CD diagrams, we focus on the case where we use the smallest total AL budget, just 5% of all examples, as this reduces human labor the most (by 95%). The raw F1-scores in this case are shown in Table 2.

As shown in Fig. 5, our Z2H significantly outperforms all non-FS rivals, as is corroborated by the raw results in Table 2. Specifically, Z2H generally performs better when paired with pre-trained FMs (especially EEGPT) than with models trained from scratch, highlighting the superiority of FMs. Also, the FMs with Z2H often have results very close to FS with just 5% of manual labeling, and in certain cases

Table 3: Online prediction F1-scores of EEGPT under an *inter-subject* setting, with total AL budget being 5% of all data and using Z2H, its ablations and rivals for fine-tuning data selection.

Method	$H1_{it}$	$H2_{it}$	$H3_{it}$	$H4_{it}$	$H5_{it}$	$H6_{it}$	$H7_{it}$	$H8_{it}$	Avg. F1	Avg. Rank (w/o FS)
FS	0.59	0.30	0.85	0.82	0.85	0.57	0.74	0.84	0.70	-
Random	0.64	0.30	**0.85**	0.80	0.80	0.49	0.70	0.83	0.68	3.50
ALSL-U	0.67	0.28	0.84	**0.81**	0.75	0.35	0.68	0.84	0.65	4.38
ALLG	0.67	0.29	**0.85**	0.79	0.60	**0.56**	0.79	0.84	0.67	3.38
ALMS	0.56	0.26	**0.85**	**0.81**	**0.83**	0.51	0.54	0.84	0.65	3.50
DUAL	0.56	0.38	**0.85**	0.80	0.61	0.45	0.65	0.81	0.64	4.62
Z2H-Only-M1	0.66	0.37	**0.85**	0.78	0.80	0.41	**0.88**	0.78	0.69	4.00
Z2H-M1-M2	0.52	0.11	0.82	0.58	0.65	0.12	0.21	0.85	0.48	7.00
Z2H (ours)	**0.72**	**0.41**	**0.85**	0.78	0.82	0.46	0.66	**0.86**	**0.70**	**2.62**

(e.g. MO3), greatly outperforms FS, thanks to the power of Z2H to select the most informative examples. Among the rival methods, Z2H-Only-M1 is generally the second best, outperforming existing UAL methods. By contrast, Z2H-M1-M2 trails behind, which seems to suggest that M2 has a negative effect. However, we now explore *why* Z2H performs well and how M2 has a positive impact on it.

Specifically, we look into H6 and MO3, and show the confusion matrices of the selected training data by all variants of Z2H and the best-performing existing UAL method when paired with EEGPT, in Fig. 6. For H6, the UAL method, ALMS, selects far fewer positive examples than all Z2H variants, hence its sub-optimal result. Z2H-Only-M1 yields better balance between the two classes, thanks to its intentional bias toward the IEDs. However, its performance is restricted by the limited size of the selected data. By contrast, Z2H-M1-M2 yields far larger data as it selects all available training data, with most of it labeled automatically in M2, yet its results are affected by both a relatively large number of mislabels and great data imbalance. Z2H, on the other hand, exploits KTEF in M3 to not only filter most mislabeled examples, but also enhance data balancing, as mentioned in Section 2.3. For MO3, DUAL and Z2H-Only-M1 have better data balance, yet are limited by the selected data size, while Z2H-M1-M2 is again affected by mislabeled data and poor data balance. By contrast, Z2H has no mislabeled data and the best trade-off between data size and data balance. *In summary, Z2H can often find the sweet spot between data size, data balance and label correctness*, This has much to do with the contribution of M2, as it enables Z2H to break the limit of the AL budget and enjoy far greater data size, which can be vital in the face of high data heterogeneity.

[Extending to an *inter*-subject setting] So far, we have focused on an *intra*-subject setting where the training and testing data come from the same subject. We now evaluate Z2H in an *inter*-patient setting where the model is trained and test on different subjects, a setting with greater real-world utility. Specifically, we apply a leave-one-out strategy for the human subjects, where we train the model on all subjects but one and use the latter for testing. Limited by computational resources, we only consider the case where the total AL budget is 5% and the core model is EEGPT (which generally performed the best in the *intra*-subject setting). The results are shown in Table 3, where Hi_{it} indicates the case where the subject Hi is used for testing. Again, Z2H generally outperforms all other methods.

3.2 Intermediate Output in Module M1

We now examine the intermediate outputs of Z2H besides the final output (i.e. the selected training data). In M1, to meet the user demand for swift data exploration, our Amp_{Dist} sampling strategy aims to identify as *many* IEDs as possible with very few user queries (e.g. 10-50). Accordingly, for a fixed, very small AL budget k, we consider the *precision at k (p@k)*, i.e. the proportion of IEDs among the first n examples queried. We let k be 10 and 50, which are very small compared to the total data size in Table 1. For rival methods, besides the existing UAL methods used in Section 3.1, we consider 3 knowledge-guided variants of Amp_{Dist}: $Dist$ which follows an NN distance ascending order, Amp which follows a P2P distance descending order with no NN distance-based skipping, and $Dist_{Amp}$ which uses NN distances to decide the primary query order, following an NN distance ascending order while skipping examples with P2P amplitudes below the median.

As shown in Table 4, random sampling performs the worst due to the rarity of IEDs. The sophisticated data-driven methods are generally outperformed by much simpler knowledge-guided ones, thanks to the power of domain knowledge. Amp and Amp_{Dist} usually perform better than $Dist$ and $Dist_{Amp}$, validating our claim in Section 2.1 that P2P amplitudes are more discriminative than NN distances. Our Amp_{Dist} outperforms Amp as NN distances can help boost the performance as an ancillary metric. Note that in some cases, the $p@k$ values drop as k increases, likely because our Amp_{Dist} can prioritize high-confidence IED examples over the rest, and a larger k can thus lead to some lower-confidence queries. Even so, Amp_{Dist} still generally outperforms its rivals when k is 50.

3.3 Intermediate Output in Module M2

In M2, to meet the user demand for offline data analysis, our ASCENSION needs to label the U set as accurately as possible, based on the initial PL set obtained from Amp_{Dist} in M1, *before* data filtering in M3. Thus, we look into the transduction F1-score in M2. First, we examine the effect of AL budget allocation between M1 and M2, considering allocating 10% : 10% : 100% of the total budget to M1 (note that it is impossible to allocate 0% of the budget to M1). Fig. 7 shows the F1-scores under different total budgets. While all other allocations yield similar results, the case 10 : 0, where 100% of the budget is allocated to M1 and ASCENSION is skipped completely, has far worse F1-scores, which highlights the enhancement power of ASCENSION. Also, Fig. 8 shows the CD diagram for statistical tests run over all combinations of 6 total AL budgets and 11 subjects, which indicates that 7 : 3, i.e., allocating 70% of the total budget to M1, leads to the best result, thus justifying this setting in previous experiments.

To further validate the posterior enhancement power of ASCENSION, we compare it against the following methods: **TC-ST-1NN, RoSAS [48], FeaWAD [50] and DeepSAD [37]**, which are state-of-the-art semi-supervised learners with TC-ST-1NN being the two-class variant of ST-1NN. Since our ASCENSION is tailored to the one-class ST-1NN, it is impossible for these methods to benefit from the posterior active enhancement of ASCENSION. To be fair to these rival methods, we allocate all the budget to M1, similar to the aforementioned 10 : 0 setting for M1 and M2. Moreover, these methods exploit both the initial PL and NL examples obtained in M1, as opposed to our one-class approach which only utilizes the former. As shown in Table 5, our ASCENSION defeats all semi-supervised baselines, again showcasing its enhancement power.

Table 4: Precisions at different numbers of user queries for Module M1

Metric	Method	H1	H2	H3	H4	H5	H6	H7	H8	MO1	MO2	MO3	Avg. $p@k$	Avg. Rank
$p@10$	Random	0.04	0.02	0.10	0.01	0.08	0.04	0.11	0.05	0.03	0.01	0.00	0.04	5.91
	ALSL-U	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.00
	ALLG	0.00	0.20	0.00	0.00	0.20	0.10	0.10	0.00	0.10	0.10	0.00	0.07	5.82
	ALMS	0.00	0.10	0.20	0.00	0.10	0.00	0.00	0.50	0.20	0.00	0.00	0.10	5.73
	DUAL	0.00	0.00	0.10	0.00	0.41	0.00	0.66	0.75	0.72	0.33	0.59	0.32	4.82
	Dist	**1.00**	0.40	**0.90**	**0.80**	**1.00**	0.00	**1.00**	0.70	0.20	0.00	0.00	0.55	2.82
	Amp	0.90	0.50	**0.90**	**0.80**	0.90	**0.70**	**1.00**	**1.00**	**1.00**	**0.90**	**1.00**	0.87	1.64
	$Dist_{Amp}$	**1.00**	0.40	**0.90**	**0.80**	**1.00**	0.00	**1.00**	0.70	0.20	0.00	0.00	0.55	2.82
	Amp_{Dist} (ours)	**1.00**	**0.60**	**0.90**	**0.80**	**1.00**	**0.70**	**1.00**	**1.00**	**1.00**	**0.90**	**1.00**	**0.90**	**1.00**
$p@50$	Random	0.02	0.02	0.11	0.03	0.05	0.02	0.09	0.04	0.06	0.01	0.00	0.04	6.55
	ALSL-U	0.00	0.04	0.02	0.00	0.02	0.04	0.02	0.02	0.02	0.00	0.02	0.02	7.27
	ALLG	0.00	0.12	0.10	0.04	0.18	0.04	0.12	0.10	0.06	0.02	0.00	0.07	5.64
	ALMS	0.00	0.04	0.14	0.04	0.14	0.02	0.08	0.18	0.18	0.00	0.00	0.07	6.00
	DUAL	0.04	0.00	0.10	0.02	0.47	0.00	0.73	0.67	0.79	0.24	0.37	0.31	5.45
	Dist	**0.98**	**0.52**	**0.82**	**0.90**	**1.00**	0.00	0.98	0.78	0.28	0.00	0.00	0.57	2.82
	Amp	0.90	0.50	0.80	0.82	0.98	0.74	0.96	0.76	**0.98**	0.82	**0.76**	0.82	2.91
	$Dist_{Amp}$	**0.98**	**0.52**	**0.82**	**0.90**	**1.00**	0.00	0.98	0.78	0.28	0.00	0.00	0.57	2.82
	Amp_{Dist} (ours)	**0.98**	**0.52**	0.72	0.80	**1.00**	**0.84**	**1.00**	**0.90**	0.88	**0.90**	0.72	**0.84**	**1.73**

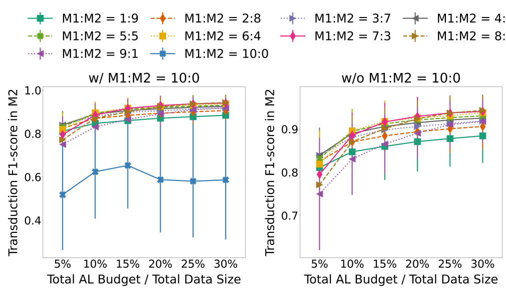


Figure 7: M2 Transduction F1-scores of varying AL budget allocation ratios between M1 and M2, averaged over all 11 human/mouse subjects. The two sub-figures are identical except that the one to the right excludes the 10:0 case for visual clarity.

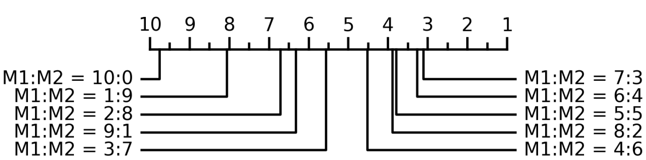


Figure 8: CD diagrams of AL budget allocations between M1 and M2. The ones to the right perform better.

Table 5: F1-scores of transduction methods for Module M2

Methods	H1	H2	H3	H4	H5	H6	H7	H8	MO1	MO2	MO3	Avg. F1	Avg. Rank
TC-ST-1NN	0.23	0.51	**0.80**	**0.78**	0.17	0.46	**0.88**	0.72	**0.87**	0.90	0.85	0.65	2.27
RoSAS	0.78	0.45	0.40	0.54	0.62	0.52	0.46	0.78	0.35	0.67	0.85	0.58	2.73
FeaWAD	0.21	0.18	0.76	0.34	0.46	0.24	0.75	0.24	0.21	0.09	0.02	0.32	3.73
DeepSAD	0.21	0.17	0.27	0.21	0.35	0.17	0.39	0.27	0.17	0.09	0.02	0.21	4.55
ASCENSION (ours)	**0.91**	**0.69**	0.72	**0.78**	**0.71**	**0.78**	0.77	**0.90**	0.70	**0.91**	**0.87**	**0.79**	**1.73**

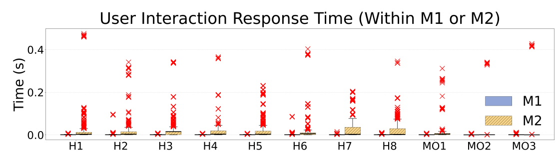


Figure 9: User interaction response times within M1 and M2 ("×" indicates outliers.)

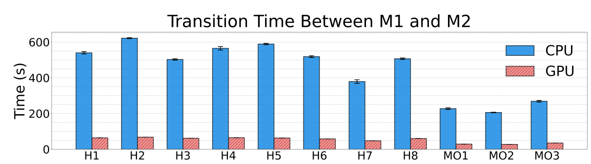


Figure 10: Transition time between M1 and M2

3.4 User Interaction Response Time

Finally, we look into the user interaction response time of Z2H. This entails inter-query delays within M1 and M2, and the transition time between M1 and M2 to run ST-1NN. The former results are shown in Fig. 9, where the delays are negligible. The latter results are shown in Fig. 10, where our GPU acceleration method can boost the efficiency by 7.4-9.4x, reducing the user waiting time to around 1 minute, ensuring (near-)real-time user interaction.

4 RELATED WORK

[IED Detection] Since 1972, various machine learning methods have been proposed for IED detection [2]. While early works drew on shallow learners with pre-defined features [3, 38], recent ones mainly adopt deep models such as CNN [4, 10, 13, 15, 34, 40, 43, 49], LSTMs [16, 29], and Transformers [35]. Some works have explored reducing manual labeling via semi-supervision [15] and data augmentation [15, 16], yet they can still require large amounts of manual labeling as they cannot select the most informative examples for manual labeling, which is the problem our Z2H addresses. Regarding data imbalance, [43] proposed a novel loss function to optimize the F_β-score for a fully labeled training set, without looking into training data selection. In the broader EEG analysis community, several foundation models [11, 21, 42] have been proposed, which can be fine-tuned for IED detection. As shown in our experiments, Z2H can effectively select fine-tuning data for such models.

[UAL] As a special case of AL, existing UAL methods typically rank the candidates by their power to reconstruct the data [20, 22–25, 28, 31, 41] or reduce prediction uncertainty [8, 41] in a latent space that can be shallow [8, 20, 24, 25, 31] or deep-learned [22, 23, 28, 41], favoring examples that can best preserve the global [8, 23–25, 28, 31, 41] or local [20, 23, 28, 41] data characteristics. Only a few methods [23, 41] explicitly handle data imbalance and heterogeneity by applying K-means to uncover diverse local clusters. However, such methods can fall short in handling chaotic data and be restricted by the limited AL budget in terms of the selected data size. Besides, they typically have limited time and (or) memory efficiency due to their heavy reliance on complex matrix operations.

5 CONCLUSIONS

In this work, we proposed Z2H, a model-agnostic, GPU-accelerated, knowledge-guided UAL pipeline that starts from a *completely unlabeled* EEG dataset to select informative examples to train or fine-tune a user-defined deep IED detection model, while producing intermediate results useful to meet diverse user demands. Effectively handling data imbalance and heterogeneity, Z2H can reduce the manual labeling needed by 95% while yielding comparable or even better results when paired with a pre-trained EEG foundation model [11, 42], which can potentially reduce the workload of neuroscientists from weeks to days or hours. It can also achieve (near-)real-time user interaction with good time and memory efficiency.

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